📘 Chapter 17 – BSOD, Servers & Active Directory (AD)

💀 1. BSOD (Blue/Black Screen of Death)

👉 In older Windows → **Blue Screen of Death (BSOD)**
👉 In newer OS → sometimes appears as **Black Screen of Death**

🧠 What is BSOD?

BSOD is a **critical system failure screen** that stops Windows immediately to prevent damage.

📌 **Real-Life Example:**

👉 Like an **MCB (Mini Circuit Breaker)** in your house ⚡

- If too much current comes → MCB trips
- Similarly → if system overload happens → BSOD appears

⚠️ Why BSOD Happens? (Very Important for Interview)

🔹 **1. RAM Overload** 🧠

Example:

- Total RAM = 16GB
- OS uses = 3.7GB
- Apps use = 14GB

👉 Total = 17.7GB ❌ (more than capacity)

👉 System retries → overload → **BSOD**

🔹 **2. Faulty RAM**

👉 Use Windows Memory Diagnostic tool:

- Checks if RAM is working properly

🔹 **3. HDD / SSD Issues** 💽

- Bad sectors
- Incorrect indexing
  👉 Causes file corruption → BSOD

🔹 **4. Driver Issues** 🚫

- Old drivers
- Incompatible drivers

👉 New OS cannot handle old driver → **Kernel Error**

🔹 **5. OS Corruption** 🧩

- System32 files damaged
- Missing files

👉 OS fails → BSOD

🔹 **6. Processor Overload** 🧠

- Heavy tasks → CPU overload
  👉 System crash

🔹 **7. SMPS (Power Supply)** ⚡

- Voltage fluctuation
  👉 Sudden shutdown → BSOD

🎯 Quick Troubleshooting for BSOD

✓ Check RAM usage
✓ Run Memory Diagnostic
✓ Update drivers (official website only)
✓ Run:

- sfc /scannow
- chkdsk
  ✓ Check power supply
  ✓ Remove recent apps

🖥️ 2. What is a Server?

👉 A **server provides services** to client machines

📌 **Example:**

- Your laptop = Client 💻
- Server = Service provider 🖥️

🌐 Types of Servers (Must Know)

Server Type	Function
🌍 Web Server	Hosts websites
🧮 App Server	Runs applications
🌐 DNS Server	Converts domain → IP
🔐 VPN Server	Secure remote access
📁 File Server	Stores & shares files
🗄️ Database Server	Stores structured data

Server Type	Function
🏢 AD Server	Manages users & systems
📧 Mail Server	Handles emails
🖥️ VMware Server	Virtualization
📡 DHCP Server	Assigns IP automatically

🏢 3. Active Directory (AD) Deep Understanding

👉 Active Directory = **Centralized Database System**

Stores:

- Users 👤
- Computers 💻
- Printers 🖨️
- Policies 🔐

🧠 Key Components

👤 User

- Individual person (employee)

🗂️ OU (Organizational Unit)

- Department/group (HR, IT, Finance)

📜 GPO (Group Policy Object)

- Rules applied to OU

🌐 Domain Controller

- Machine where AD is installed

⚙️ How AD Works (Real Flow)

1. Admin creates user 👤
2. Sets username & password 🔑
3. Manager shares credentials
4. User logs in → forced to change password 🔁
5. GPO automatically applied 🎯

🔄 If Policy Not Applied?

👉 Run Command Prompt (Admin):

gpupdate /force

👉 Forces system to connect with Domain Controller

🔗 4. Domain Join Methods

✅ Method 1: System Properties

1. Right-click **This PC → Properties**
2. Click **Change Settings**
3. Click **Domain**
4. Enter domain name
5. Enter admin credentials

✅ Method 2: Settings

1. Go to:

Settings → Accounts → Access Work or School

2. Click **Join this device to a domain**

3. Enter domain ID

📌 **Requirements:**

- Same network (Intranet) 🌐
- OR VPN if remote 🔐

🌍 5. VPN Types

🔹 **Site-to-Site VPN**

- Connects **office to office**
- Example:
  - Mumbai ↔ Pune

👉 Secure tunnel between routers

🔹 **Remote Access VPN**

- Connects **user to office**

👉 Tools:

- Cisco AnyConnect
- GlobalProtect

🛠️ 6. PRACTICAL TASK – FULL STEP-BY-STEP 🔥

🧰 PART 1: Install Active Directory

Step 1:

- Open **Server Manager**

Step 2:

- Click **Add Roles and Features**

Step 3:

- Select:
  👉 **Active Directory Domain Services (AD DS)**

Step 4:

- Install & Restart

Step 5:

- Click:
  👉 **Promote this server to Domain Controller**

Step 6:

- Select:
  👉 Add new forest → enter:

company.local

🗂️ PART 2: Create OU (4 OUs)

1. Open:
   👉 Active Directory Users & Computers
2. Right-click domain → New → Organizational Unit
3. Create:

- IT

- HR
- Finance
- Marketing

👤 PART 3: Create 20 Users

1. Right-click OU → New → User
2. Enter:

- Name
- Username
- Password

👉 Create **5 users in each OU**

📦 PART 4: Create GPO (Block Control Panel)

Step 1:

- Open:
  👉 Group Policy Management

Step 2:

- Right-click OU → Create GPO

👉 Name:

Block_Control_Panel

Step 3:

- Edit GPO:

Go to:

User Configuration → Administrative Templates → Control Panel

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Step 4:

- Enable:
  👉 **Prohibit access to Control Panel and PC Settings**

🔗 PART 5: Link GPO

- Right-click OU → Link Existing GPO
- Select created GPO

💻 PART 6: Join Client Machine to Domain

Step 1:

- Connect to same network / VPN

Step 2:

- Go to:

System Properties → Change Settings

Step 3:

- Enter domain:

company.local

Step 4:

- Enter admin credentials

Step 5:

- Restart system 🔁

✅ Verification:

Login with domain user:

👉 Try opening Control Panel
❌ It should be blocked

🎯 Final Troubleshooting Tips 💡

✓ If GPO not working → run gpupdate /force
✓ Check network connectivity
✓ Verify domain join
✓ Restart system

🧾 Final Summary

- BSOD = **System Protection Mechanism** 💀
- Servers = **Service Providers** 🖥️
- AD = **Centralized Control System** 🏢
- GPO = **Rules Engine** 🔐
- Domain = **Network Identity** 🌐

📘 Q & A

💀 SECTION 1: BSOD (Blue Screen of Death) – Interview Questions

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❓ Q1: What is BSOD and why does it occur?

✅ **Answer:**

BSOD (Blue Screen of Death) is a **critical system error screen** in Windows that appears when the operating system encounters a **fatal error it cannot recover from**.

👉 It immediately stops the system to **prevent hardware damage or data corruption**.

🧠 **Simple Explanation:**

Like an **MCB trip** ⚡ — system shuts down to protect itself.

🔍 **Common Causes:**

- Faulty RAM 🧠
- Driver issues 🚫
- OS corruption 📁
- Disk errors 💽
- Overheating 🔥
- Power supply issues ⚡

❓ Q2: How do you troubleshoot a BSOD issue step-by-step?

✅ **Answer (Professional Approach):**

🔧 **Step-by-Step Troubleshooting:**

1️⃣ **Note Error Code**

- Example: IRQL_NOT_LESS_OR_EQUAL

- Helps identify root cause

2️⃣ Boot into Safe Mode

- To isolate driver/software issue

3️⃣ Check Recent Changes

- New driver or software install?

4️⃣ Run System File Check

sfc /scannow

5️⃣ Run Disk Check

chkdsk /f /r

6️⃣ Update Drivers

- Especially GPU, chipset, network

7️⃣ Check RAM

- Use Windows Memory Diagnostic

8️⃣ Check Event Viewer

- Analyze crash logs 📊

9️⃣ Check Temperature

- Overheating causes shutdown

🔟 Last Option

- System Restore or OS Repair

❓ Q3: What tools are used to analyze BSOD?

✅ Answer:

- Event Viewer 📊
- Windows Memory Diagnostic 🧠
- BlueScreenView 🔍

- WinDbg (Advanced debugging tool)

❓ **Q4: What is the difference between hardware BSOD and software BSOD?**

✅ **Answer:**

Type	Cause	Example
Hardware BSOD	Physical issue	Faulty RAM
Software BSOD	OS/driver issue	Bad driver

👉 Hardware = Replace component
👉 Software = Fix OS/driver

❓ **Q5: What is MiniDump file?**

✅ **Answer:**

MiniDump is a **small crash log file** created during BSOD.

📍 Location:

C:\Windows\Minidump

👉 Used for **debugging crash reasons**

🖥️ **SECTION 2: SERVER – Interview Questions**

❓ **Q6: What is a server?**

✅ **Answer:**

A server is a **computer or system that provides services to other computers (clients)** over a network.

🧠 **Example:**

- Laptop = Client 💻

- Server = Service provider 🖥️

❓ Q7: What are different types of servers?

✅ **Answer:**

Server Type	**Function**
Web Server 🌍	Hosts websites
File Server 📁	Stores files
DNS Server 🌐	Domain → IP
DHCP Server 📡	Assigns IP
Mail Server 📧	Email services
AD Server 🏢	User management
Database Server 🗄️	Stores data

❓ Q8: What is the difference between client and server?

✅ **Answer:**

Client 💻	**Server** 🖥️
Requests service	Provides service
End user machine	Backend system

❓ Q9: What is DHCP?

✅ **Answer:**

DHCP (Dynamic Host Configuration Protocol) automatically assigns:

- IP address 🌐
- Subnet mask

- Gateway

👉 Without DHCP → manual IP needed

❓ **Q10: What is DNS?**

✅ **Answer:**

DNS converts **domain names into IP addresses**

👉 Example:

google.com → 142.x.x.x

🏢 **SECTION 3: ACTIVE DIRECTORY (AD) – Interview Questions**

❓ **Q11: What is Active Directory?**

✅ **Answer:**

Active Directory is a **centralized directory service by Microsoft** used to manage:

- Users 👤
- Computers 💻
- Policies 🔐
- Resources 📁

❓ **Q12: What is OU (Organizational Unit)?**

✅ **Answer:**

OU is a **container inside Active Directory** used to organize:

- Departments
- Users
- Systems

👉 Helps in applying policies easily

❓ Q13: What is Group Policy (GPO)?

✅ **Answer:**

GPO is used to **apply rules/settings to users or computers centrally**.

🎯 **Example:**

- Disable Control Panel 🚫
- Set company wallpaper 🖼️

❓ Q14: What is a Domain Controller?

✅ **Answer:**

A Domain Controller is a **server that runs Active Directory services** and handles:

- Authentication 🔐
- Authorization 🚫

❓ Q15: What is Authentication vs Authorization?

✅ **Answer:**

Concept	Meaning
Authentication	Who are you? 🔐
Authorization	What can you access? 🚫

❓ Q16: What is Domain Join?

✅ **Answer:**

Connecting a computer to a domain so it can:

- Follow company policies
- Use centralized login

❓ Q17: What is gpupdate /force?

✅ **Answer:**

Command used to **force apply Group Policy immediately**

gpupdate /force

⚙️ SECTION 4: PRACTICAL / JOB-ORIENTED QUESTIONS

❓ Q18: GPO is not applying. What will you do?

✅ **Answer (Stepwise):**

1️⃣ Check system is domain joined
2️⃣ Check OU is correct
3️⃣ Verify GPO is linked
4️⃣ Run:

gpupdate /force

5️⃣ Run:

gpresult /r

6️⃣ Restart system

❓ **Q19: User cannot login to domain. How will you fix?**

✅ **Answer:**

1️⃣ Check network connectivity 🌐
2️⃣ Verify domain server reachable
3️⃣ Check user account status
4️⃣ Reset password 🔑
5️⃣ Check time sync ⏰
6️⃣ Check DNS configuration

❓ **Q20: How do you create a user in AD?**

✅ **Answer:**

1️⃣ Open AD Users & Computers
2️⃣ Right-click OU
3️⃣ Click → New → User
4️⃣ Enter details
5️⃣ Set password
6️⃣ Enable account

❓ **Q21: What is onboarding and offboarding?**

✅ **Answer:**

Process	**Meaning**
Onboarding 👤	New user creation
Offboarding 🚪	Disable/delete user

❓ **Q22: How do you block Control Panel using GPO?**

✅ **Answer:**

1️⃣ Open Group Policy Management
2️⃣ Create GPO
3️⃣ Go to:
User Config → Admin Templates → Control Panel
4️⃣ Enable:
👉 Prohibit access

❓ **Q23: What are prerequisites for domain join?**

✅ **Answer:**

- Same network 🌐
- Proper DNS
- Domain credentials 🔐

🔐 **SECTION 5: SECURITY & REAL SCENARIOS**

❓ **Q24: Why should admin rights be limited?**

✅ **Answer:**

- Prevent malware spread 🦠
- Reduce risk of system damage
- Improve security 🔐

❓ Q25: What happens if GPO is misconfigured?

✅ **Answer:**

- Users lose access
- System restrictions applied wrongly
- Security issues

❓ Q26: How does AD improve security?

✅ **Answer:**

- Central control
- Policy enforcement
- Access restriction
- Logging & auditing

🧾 FINAL SUMMARY

Concept	Meaning
BSOD 💀	System crash protection
Server 🖥️	Service provider
AD 🏢	Central management
OU 📁	Grouping system
GPO 🔐	Rule engine
Domain 🌐	Network identity
